$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{r@{\,:\,}l} \mathsf{state} & Loc \times Loc \times Loc \times int \times int \times int \times int \times int \Rightarrow State \\ \mathsf{noncrit} & Loc \\ \mathsf{level1} & Loc \\ \mathsf{level2} & Loc \\ \mathsf{crit} & Loc \\ \mathsf{success} & State \\ \mathsf{error} & State \end{array} \right\}$$

$$
\left\{
\begin{array}{rrll}
1: & \mathsf{state}(\mathsf{noncrit}, \mathit{proc1}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \to & \mathsf{state}(\mathsf{level1}, \mathit{proc1}, \mathit{proc2}, \mathit{p01}, \mathit{p1}, \mathit{p2}, \mathit{y11}, \mathit{y2}) & [\,\mathit{p01} = 1 \wedge \mathit{y11} = 0\,] \\
2: & \mathsf{state}(\mathsf{level1}, \mathit{proc1}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \to & \mathsf{state}(\mathsf{level2}, \mathit{proc1}, \mathit{proc2}, \mathit{p01}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y21}) & [\,(\mathit{y1} \neq 0 \vee ((\mathit{p1} < 1) \wedge (\mathit{p2} < 1))) \wedge \mathit{p01} = 2 \wedge \mathit{y21} = 0\,] \\
3: & \mathsf{state}(\mathsf{level2}, \mathit{proc1}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \to & \mathsf{state}(\mathsf{crit}, \mathit{proc1}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) & [\,\mathit{y2} \neq 0 \vee ((\mathit{p1} < 2) \wedge (\mathit{p2} < 2))\,] \\
4: & \mathsf{state}(\mathsf{crit}, \mathit{proc1}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \to & \mathsf{state}(\mathsf{noncrit}, \mathit{proc1}, \mathit{proc2}, \mathit{p01}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) & [\,\mathit{p01} = 0\,] \\
5: & \mathsf{state}(\mathit{proc0}, \mathsf{noncrit}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \to & \mathsf{state}(\mathit{proc0}, \mathsf{level1}, \mathit{proc2}, \mathit{p0}, \mathit{p11}, \mathit{p2}, \mathit{y11}, \mathit{y2}) & [\,\mathit{p11} = 1 \wedge \mathit{y11} = 1\,] \\
6: & \mathsf{state}(\mathit{proc0}, \mathsf{level1}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \to & \mathsf{state}(\mathit{proc0}, \mathsf{level2}, \mathit{proc2}, \mathit{p0}, \mathit{p11}, \mathit{p2}, \mathit{y1}, \mathit{y21}) & [\,(\mathit{y1} \neq 1 \vee ((\mathit{p0} < 1) \wedge (\mathit{p2} < 1))) \wedge \mathit{p11} = 2 \wedge \mathit{y21} = 1\,] \\
7: & \mathsf{state}(\mathit{proc0}, \mathsf{level2}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \to & \mathsf{state}(\mathit{proc0}, \mathsf{crit}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) & [\,\mathit{y2} \neq 1 \vee ((\mathit{p0} < 2) \wedge (\mathit{p2} < 2))\,] \\
8: & \mathsf{state}(\mathit{proc0}, \mathsf{crit}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \to & \mathsf{state}(\mathit{proc0}, \mathsf{noncrit}, \mathit{proc2}, \mathit{p0}, \mathit{p11}, \mathit{p2}, \mathit{y1}, \mathit{y2}) & [\,\mathit{p11} = 0\,] \\
9: & \mathsf{state}(\mathit{proc0}, \mathit{proc1}, \mathsf{noncrit}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \to & \mathsf{state}(\mathit{proc0}, \mathit{proc1}, \mathsf{level1}, \mathit{p0}, \mathit{p1}, \mathit{p21}, \mathit{y11}, \mathit{y2}) & [\,\mathit{p21} = 1 \wedge \mathit{y11} = 2\,] \\
10: & \mathsf{state}(\mathit{proc0}, \mathit{proc1}, \mathsf{level1}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \to & \mathsf{state}(\mathit{proc0}, \mathit{proc1}, \mathsf{level2}, \mathit{p0}, \mathit{p1}, \mathit{p21}, \mathit{y1}, \mathit{y21}) & [\,(\mathit{y1} \neq 2 \vee ((\mathit{p0} < 1) \wedge (\mathit{p1} < 1))) \wedge \mathit{p21} = 2 \wedge \mathit{y21} = 2\,] \\
11: & \mathsf{state}(\mathit{proc0}, \mathit{proc1}, \mathsf{level2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \to & \mathsf{state}(\mathit{proc0}, \mathit{proc1}, \mathsf{crit}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) & [\,\mathit{y2} \neq 2 \vee ((\mathit{p0} < 2) \wedge (\mathit{p1} < 2))\,] \\
12: & \mathsf{state}(\mathit{proc0}, \mathit{proc1}, \mathsf{crit}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \to & \mathsf{state}(\mathit{proc0}, \mathit{proc1}, \mathsf{noncrit}, \mathit{p0}, \mathit{p1}, \mathit{p21}, \mathit{y1}, \mathit{y2}) & [\,\mathit{p21} = 0\,] \\
13: & \mathsf{state}(\mathit{proc0}, \mathit{proc1}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \to & \mathsf{success} & \\
14: & \mathsf{state}(\mathsf{crit}, \mathsf{crit}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \to & \mathsf{error} & \\
15: & \mathsf{state}(\mathsf{crit}, \mathit{proc1}, \mathsf{crit}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \to & \mathsf{error} & \\
16: & \mathsf{state}(\mathit{proc0}, \mathsf{crit}, \mathsf{crit}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \to & \mathsf{error} &
\end{array}
\right\}
$$

$$\left\{ \quad 17: \quad \mathsf{state}(\mathsf{noncrit}, \mathsf{noncrit}, \mathsf{noncrit}, 0, 0, 0, y1, y2) \Rightarrow \mathsf{success} \quad \right\}$$

$$\Sigma = \Sigma^{int}_{theory} \cup \left\{ \begin{array}{rl} \mathsf{state} : & Loc \times Loc \times Loc \times bitvec_2 \times bitvec_2 \times bitvec_2 \times bitvec_2 \times bitvec_2 \Rightarrow State \\ \mathsf{noncrit} : & Loc \\ \mathsf{level1} : & Loc \\ \mathsf{level2} : & Loc \\ \mathsf{crit} : & Loc \\ \mathsf{success} : & State \\ \mathsf{error} : & State \end{array} \right\}$$

$$
\left\{
\begin{array}{rrl}
1: & \mathsf{state}(\mathsf{noncrit}, \mathit{proc1}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \rightarrow \mathsf{state}(\mathsf{level1}, \mathit{proc1}, \mathit{proc2}, \mathit{p01}, \mathit{p1}, \mathit{p2}, \mathit{y11}, \mathit{y2}) & [\,\mathit{p01} = \mathsf{\#b01} \wedge \mathit{y11} = \mathsf{\#b00}\,] \\
2: & \mathsf{state}(\mathsf{level1}, \mathit{proc1}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \rightarrow \mathsf{state}(\mathsf{level2}, \mathit{proc1}, \mathit{proc2}, \mathit{p01}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y21}) & [\,(\mathit{y1} \neq \mathsf{\#b00} \vee ((\mathit{p1}\,\mathsf{bvult\#b01}) \wedge (\mathit{p2}\,\mathsf{bvult\#b01}))) \wedge \mathit{p01} = \mathsf{\#b10} \wedge \mathit{y21} = \mathsf{\#b00}\,] \\
3: & \mathsf{state}(\mathsf{level2}, \mathit{proc1}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \rightarrow \mathsf{state}(\mathsf{crit}, \mathit{proc1}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) & [\,\mathit{y2} \neq \mathsf{\#b00} \vee ((\mathit{p1}\,\mathsf{bvult\#b10}) \wedge (\mathit{p2}\,\mathsf{bvult\#b10}))\,] \\
4: & \mathsf{state}(\mathsf{crit}, \mathit{proc1}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \rightarrow \mathsf{state}(\mathsf{noncrit}, \mathit{proc1}, \mathit{proc2}, \mathit{p01}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) & [\,\mathit{p01} = \mathsf{\#b00}\,] \\
5: & \mathsf{state}(\mathit{proc0}, \mathsf{noncrit}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \rightarrow \mathsf{state}(\mathit{proc0}, \mathsf{level1}, \mathit{proc2}, \mathit{p0}, \mathit{p11}, \mathit{p2}, \mathit{y11}, \mathit{y2}) & [\,\mathit{p11} = \mathsf{\#b01} \wedge \mathit{y11} = \mathsf{\#b01}\,] \\
6: & \mathsf{state}(\mathit{proc0}, \mathsf{level1}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \rightarrow \mathsf{state}(\mathit{proc0}, \mathsf{level2}, \mathit{proc2}, \mathit{p0}, \mathit{p11}, \mathit{p2}, \mathit{y1}, \mathit{y21}) & [\,(\mathit{y1} \neq \mathsf{\#b01} \vee ((\mathit{p0}\,\mathsf{bvult\#b01}) \wedge (\mathit{p2}\,\mathsf{bvult\#b01}))) \wedge \mathit{p11} = \mathsf{\#b10} \wedge \mathit{y21} = \mathsf{\#b01}\,] \\
7: & \mathsf{state}(\mathit{proc0}, \mathsf{level2}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \rightarrow \mathsf{state}(\mathit{proc0}, \mathsf{crit}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) & [\,\mathit{y2} \neq \mathsf{\#b01} \vee ((\mathit{p0}\,\mathsf{bvult\#b10}) \wedge (\mathit{p2}\,\mathsf{bvult\#b10}))\,] \\
8: & \mathsf{state}(\mathit{proc0}, \mathsf{crit}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \rightarrow \mathsf{state}(\mathit{proc0}, \mathsf{noncrit}, \mathit{proc2}, \mathit{p0}, \mathit{p11}, \mathit{p2}, \mathit{y1}, \mathit{y2}) & [\,\mathit{p11} = \mathsf{\#b00}\,] \\
9: & \mathsf{state}(\mathit{proc0}, \mathit{proc1}, \mathsf{noncrit}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \rightarrow \mathsf{state}(\mathit{proc0}, \mathit{proc1}, \mathsf{level1}, \mathit{p0}, \mathit{p1}, \mathit{p21}, \mathit{y11}, \mathit{y2}) & [\,\mathit{p21} = \mathsf{\#b01} \wedge \mathit{y11} = \mathsf{\#b10}\,] \\
10: & \mathsf{state}(\mathit{proc0}, \mathit{proc1}, \mathsf{level1}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \rightarrow \mathsf{state}(\mathit{proc0}, \mathit{proc1}, \mathsf{level2}, \mathit{p0}, \mathit{p1}, \mathit{p21}, \mathit{y1}, \mathit{y21}) & [\,(\mathit{y1} \neq \mathsf{\#b10} \vee ((\mathit{p0}\,\mathsf{bvult\#b01}) \wedge (\mathit{p1}\,\mathsf{bvult\#b01}))) \wedge \mathit{p21} = \mathsf{\#b10} \wedge \mathit{y21} = \mathsf{\#b10}\,] \\
11: & \mathsf{state}(\mathit{proc0}, \mathit{proc1}, \mathsf{level2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \rightarrow \mathsf{state}(\mathit{proc0}, \mathit{proc1}, \mathsf{crit}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) & [\,\mathit{y2} \neq \mathsf{\#b10} \vee ((\mathit{p0}\,\mathsf{bvult\#b10}) \wedge (\mathit{p1}\,\mathsf{bvult\#b10}))\,] \\
12: & \mathsf{state}(\mathit{proc0}, \mathit{proc1}, \mathsf{crit}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \rightarrow \mathsf{state}(\mathit{proc0}, \mathit{proc1}, \mathsf{noncrit}, \mathit{p0}, \mathit{p1}, \mathit{p21}, \mathit{y1}, \mathit{y2}) & [\,\mathit{p21} = \mathsf{\#b00}\,] \\
13: & \mathsf{state}(\mathit{proc0}, \mathit{proc1}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \rightarrow \mathsf{success} & \\
14: & \mathsf{state}(\mathsf{crit}, \mathsf{crit}, \mathit{proc2}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \rightarrow \mathsf{error} & \\
15: & \mathsf{state}(\mathsf{crit}, \mathit{proc1}, \mathsf{crit}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \rightarrow \mathsf{error} & \\
16: & \mathsf{state}(\mathit{proc0}, \mathsf{crit}, \mathsf{crit}, \mathit{p0}, \mathit{p1}, \mathit{p2}, \mathit{y1}, \mathit{y2}) \rightarrow \mathsf{error} &
\end{array}
\right\}
$$

$$\{\ \ 17: \quad \mathsf{state}(\mathsf{noncrit}, \mathsf{noncrit}, \mathsf{noncrit}, \#\mathsf{b}00, \#\mathsf{b}00, \#\mathsf{b}00, \mathit{y1}, \mathit{y2}) \Rightarrow \mathsf{success}\ \ \}$$